Order of Operations

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS order of operations · GEMA · grouping · brackets · exponent · operation

A Multiplication and division first ●○○ ANSWER ON THE SHEET

⚠ **Watch out:** Do not just work from left to right. Multiply and divide before you add and subtract: $3 + 4 \times 5 = 3 + 20 = 23$.

1. Work out.

a) $3 + 4 \times 5 =$ _____ **b)** $20 - 12 \div 4 =$ _____ **c)** $6 \times 2 + 7 =$ _____ **d)** $18 \div 3 - 2 =$ _____

e) $10 - 2 \times 3 =$ _____ **f)** $4 + 16 \div 8 =$ _____ **g)** $5 \times 3 + 2 \times 4 =$ _____ **h)** $30 - 5 \times 5 =$ _____

i) $7 + 21 \div 7 =$ _____ **j)** $9 \times 2 - 12 \div 4 =$ _____ **k)** $40 \div 8 + 6 \times 2 =$ _____ **l)** $15 - 3 \times 4 + 6 =$ _____

B Left to right ●○○ ANSWER ON THE SHEET

Remember: Adding and subtracting are done in order from left to right. So are multiplying and dividing: $20 - 8 + 5 = 12 + 5 = 17$.

2. Work out.

a) $20 - 8 + 5 =$ _____ **b)** $15 + 4 - 9 =$ _____ **c)** $24 \div 4 \times 2 =$ _____ **d)** $6 \times 5 \div 3 =$ _____

e) $30 - 12 - 6 =$ _____ **f)** $40 \div 5 \div 2 =$ _____ **g)** $18 - 9 + 7 - 3 =$ _____ **h)** $2 \times 9 \div 6 =$ _____

C Brackets ●●○ ANSWER ON THE SHEET

Remember: GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

3. Work out.

a) $(3 + 4) \times 5 =$ _____ **b)** $18 \div (2 + 4) =$ _____ **c)** $(10 - 4) \times 3 =$ _____

d) $2 \times (9 - 5) =$ _____ **e)** $(8 + 12) \div 5 =$ _____ **f)** $30 - (7 + 8) =$ _____

g) $(6 - 2) \times (3 + 4) =$ _____ **h)** $5 + 2 \times (8 - 3) =$ _____ **i)** $(15 - 3) \div (1 + 3) =$ _____

j) $4 \times (2 + 5) - 8 =$ _____ **k)** $50 \div (12 - 2) + 7 =$ _____ **l)** $(20 - 2 \times 5) \div 2 =$ _____

D Exponents ●●● WORK IN YOUR BOOK

⚠ **Watch out:** Work out an exponent before you multiply: $2 \times 3^2 = 2 \times 9 = 18$, not 6^2.

4. Work out. Show your first step.

a) 2×3^2 **b)** $4^2 - 3 \times 5$ **c)** $5 + 2^3$

d) $(6 + 2)^2 \div 4$ **e)** $50 - 5^2$ **f)** $3^2 + 4^2$

g) $(3 + 4)^2$ **h)** $10^2 \div (2 + 3)$ **i)** $2^3 \times (7 - 4)$

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E Mixed practice ●●● — ANSWER ON THE SHEET

5. Work out.

a) $2 + 3 \times 4 - 5$ = _____ **b)** $(2 + 3) \times (4 - 1)$ = _____ **c)** $6^2 \div 4 + 1$ = _____

d) $100 - 5 \times (2 + 6)$ = _____ **e)** $3 \times 2^2 + 10$ = _____ **f)** $48 \div (2 \times 4)$ = _____

g) $(7 - 3)^2 + 5$ = _____ **h)** $8 + 12 \div 4 \times 3$ = _____ **i)** $5 \times (10 - 2 \times 3)$ = _____

j) $2^3 + 3^2 - 1$ = _____ **k)** $60 \div 5 \div 2$ = _____ **l)** $4 \times 5 - 3 \times 6$ = _____

F True or false? ●●● — WORK IN YOUR BOOK

6. A student wrote these answers. Say whether each one is right. If it is wrong, give the correct answer.

a) $4 + 6 \times 2 = 20$

b) $(4 + 6) \times 2 = 20$

c) $20 - 8 \div 4 = 3$

d) $3 \times 2^2 = 12$

e) $24 \div 4 + 2 = 4$

f) $10 - 2 + 5 = 13$

G Problems ●●● — WORK IN YOUR BOOK

7. Write one calculation for each problem, then work it out.

a) Mia buys 3 pies at $5 each and one drink for $4. How much does she spend?

b) Tane has $50. He buys 6 tickets at $7 each. How much money does he have left?

c) One class has 24 students and another has 18. All the students are put into 6 equal teams. How many are in each team?

d) A shop sells 4 boxes of 12 eggs and 3 boxes of 8 eggs. How many eggs is that altogether?

★ Extension — FOR EARLY FINISHERS

8. Put one pair of brackets into each calculation to make it true.

a) $3 + 4 \times 5 = 35$ **b)** $20 - 12 \div 4 + 2 = 18$ **c)** $2 \times 5 - 1 + 3 = 11$

9. Use the numbers 2, 3 and 5 once each, with any operations and brackets, to make 25.

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Order of Operations

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS order of operations · GEMA · grouping · brackets · exponent · operation

A Multiplication and division first ●○○ — ANSWER ON THE SHEET

⚠ **Watch out:** Do not just work from left to right. Multiply and divide before you add and subtract: $3 + 4 \times 5 = 3 + 20 = 23$.

1. Work out.

a) $2 + 3 \times 6 =$ _____ **b)** $30 - 18 \div 6 =$ _____ **c)** $5 \times 3 + 8 =$ _____ **d)** $24 \div 4 - 3 =$ _____

e) $12 - 2 \times 4 =$ _____ **f)** $5 + 20 \div 10 =$ _____ **g)** $4 \times 3 + 3 \times 5 =$ _____ **h)** $40 - 6 \times 6 =$ _____

i) $8 + 24 \div 8 =$ _____ **j)** $8 \times 2 - 15 \div 3 =$ _____ **k)** $36 \div 9 + 7 \times 2 =$ _____ **l)** $20 - 4 \times 4 + 5 =$ _____

B Left to right ●○○ — ANSWER ON THE SHEET

💡 **Remember:** Adding and subtracting are done in order from left to right. So are multiplying and dividing: $20 - 8 + 5 = 12 + 5 = 17$.

2. Work out.

a) $30 - 9 + 4 =$ _____ **b)** $18 + 6 - 8 =$ _____ **c)** $36 \div 6 \times 2 =$ _____ **d)** $8 \times 3 \div 4 =$ _____

e) $40 - 15 - 5 =$ _____ **f)** $60 \div 5 \div 3 =$ _____ **g)** $20 - 8 + 6 - 4 =$ _____ **h)** $3 \times 8 \div 6 =$ _____

C Brackets ●●○ — ANSWER ON THE SHEET

💡 **Remember:** GEMA: Grouping (brackets) first, then Exponents, then Multiplication and division, then Addition and subtraction.

3. Work out.

a) $(2 + 3) \times 6 =$ _____ **b)** $24 \div (3 + 5) =$ _____ **c)** $(12 - 5) \times 4 =$ _____

d) $3 \times (8 - 2) =$ _____ **e)** $(9 + 15) \div 6 =$ _____ **f)** $40 - (9 + 6) =$ _____

g) $(8 - 5) \times (3 + 4) =$ _____ **h)** $4 + 3 \times (9 - 5) =$ _____ **i)** $(20 - 2) \div (4 + 2) =$ _____

j) $5 \times (1 + 3) - 9 =$ _____ **k)** $60 \div (15 - 3) + 8 =$ _____ **l)** $(30 - 4 \times 3) \div 3 =$ _____

D Exponents ●●● — WORK IN YOUR BOOK

⚠ **Watch out:** Work out an exponent before you multiply: $2 \times 3^2 = 2 \times 9 = 18$, not 6^2.

4. Work out. Show your first step.

a) 3×2^3 **b)** $5^2 - 4 \times 3$ **c)** $6 + 3^2$

d) $(5 + 1)^2 \div 9$ **e)** $40 - 6^2$ **f)** $6^2 + 8^2$

g) $(2 + 6)^2$ **h)** $10^2 \div (1 + 4)$ **i)** $3^2 \times (9 - 5)$

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E Mixed practice ●●● ANSWER ON THE SHEET

5. Work out.

a) $3 + 2 \times 5 - 4$ = _____ **b)** $(3 + 4) \times (5 - 2)$ = _____ **c)** $8^2 \div 4 + 3$ = _____

d) $100 - 4 \times (3 + 7)$ = _____ **e)** $2 \times 3^2 + 20$ = _____ **f)** $54 \div (3 \times 3)$ = _____

g) $(9 - 4)^2 + 6$ = _____ **h)** $6 + 16 \div 4 \times 2$ = _____ **i)** $4 \times (12 - 3 \times 3)$ = _____

j) $3^2 + 2^4 - 1$ = _____ **k)** $80 \div 4 \div 2$ = _____ **l)** $5 \times 6 - 4 \times 7$ = _____

F True or false? ●●● WORK IN YOUR BOOK

6. A student wrote these answers. Say whether each one is right. If it is wrong, give the correct answer.

a) $5 + 3 \times 4 = 32$

b) $(5 + 3) \times 4 = 32$

c) $30 - 12 \div 6 = 3$

d) $2 \times 3^2 = 18$

e) $36 \div 6 + 3 = 4$

f) $12 - 4 + 3 = 11$

G Problems ●●● WORK IN YOUR BOOK

7. Write one calculation for each problem, then work it out.

a) Mia buys 4 pies at \$6 each and one drink for \$3. How much does she spend?

b) Tane has \$100. He buys 8 tickets at \$9 each. How much money does he have left?

c) One class has 35 students and another has 21. All the students are put into 7 equal teams. How many are in each team?

d) A shop sells 5 boxes of 10 eggs and 2 boxes of 15 eggs. How many eggs is that altogether?

★ Extension FOR EARLY FINISHERS

8. Put one pair of brackets into each calculation to make it true.

a) $2 + 6 \times 4 = 32$ **b)** $30 - 18 \div 6 + 3 = 28$ **c)** $3 \times 4 - 2 + 5 = 11$

9. Use the numbers 2, 4 and 6 once each, with any operations and brackets, to make 36.

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